



Solar-powered jet airliner

This futuristic three-storey solar powered jet is a concept airliner from designer Oscar Viñals
<http://digit.in/SolarPowJet>

Corrigendum

The Space Age story "Ice Fishing on Europa" published in our March 2015 issue should have been credited to Samir Alam. We deeply regret the error.

Space Age

and impossible events and things, which could have only been dreamed up or invented with pure thought. However, the need to achieve these impossible dreams and visions has many times defined the nature of science and technology. The sum total of popular culture; be it movies, books or even philosophy, has played a key role in innovations which are simply evidence of science catching up with imagination.

The biggest of these examples in space research is from science fiction stories in movies, television shows and books. While there are too many of these to list out, we shall hit upon the most profound influences that have shaped modern space research. The first of which is the impact of a fantastical 19th century story on the mind of one of history's finest scientific minds. Robert Goddard, the first American to build a liquid-fueled rocket, was inspired by H. G. Wells' novel War of the Worlds. The 1898 publication was a work of pure imagination that spoke of interplanetary travel. But to Goddard it was a goal worth pursuing, which he did when he successfully launched his rocket on 16 March 1926 paving the way for all future space flights.

Another critical technology - atomic power - was also inspired by H.G. Wells' plentiful imagination, and came from his novel The World Set Free. This story was written in 1914 and explored the idea of artificial atomic energy. The world of the story was set in 1933 and was almost prophetic since physicist Leo Szilard read



Astronaut Sunita Williams ran in the Boston Marathon from the ISS as imagined in the movie

the book and was inspired to solve the nuclear chain equation by 1933 giving birth to the nuclear age of energy manufacturing and war.

However, the biggest impact on space research has most likely come from movies. The most influential of which has been the epic cinematic story written in

1968 by Arthur C. Clarke and directed by Stanley Kubrick – 2001: A Space Odyssey. In terms of real innovations, this movie has for many space scientists served as the inspiration of how practical space exploration might be in the near future. The visual story elements from the movie such as the manned orbiting station, were mainly responsible for giving birth to the International Space Station. The film shaped many of the realities that would soon come to pass as fact, such as low Earth orbits, permanent crews, flat screen displays, glass cockpits, entertainment media in orbit and human athletics in orbit.

In fact, this inspiration was commemorated on the 40th anniversary of the film, when in 2008 NASA sent a direct message from the International Space Station to a special screening of the film at the Academy of Motion Pictures Arts and Sciences in California.

Video Games. Really.

An unlikely source if ever there was one, but video games are now potentially becoming a source for inspiration for discovering innovations for space research. While institutions like NASA have always supported games, as in the case of the uber-realistic Moonbase Alpha and Mars Rover Landings, this has mostly been an educational outreach and marketing. But with 2013's Dark Side of the Jam and the Night Rover Challenge during the Centennial Challenges initiative, NASA is challenging game developers to address real problems and provide innovations. The Night Rover Challenge itself was used to explore innovations in energy storage technologies.

However, it must be said that the results of this radically new experience has yet to be released as something significant. After all, how can video game developers shed new light on something as niche as energy storage in space? But as history has shown – inspiration can come from anywhere – be it dreams of cows or traffic cameras, futuristic movies or fantastical story books, we must never underestimate the ability of the human mind to find an epiphany in the ether of life. And with it to further our own quest towards the stars. 



An image from 2001: A Space Odyssey's international rotating wheel space station